

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
library(tidyverse)
library(lubridate)
library(here)

here()
```

```
## [1] "/home/guest/ENV 872/EDE_Fall2025"
```

```
getwd()
```

```
## [1] "/home/guest/ENV 872/EDE_Fall2025"
```

```
Neonics <- read.csv(  
  file = here("Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),  
  stringsAsFactors = TRUE,  
)  
  
Litter <- read.csv(  
  file = here("Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),  
  stringsAsFactors = TRUE,  
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Due to their widespread use, the effect of neonicotinoids is not just simply felt as an insecticide, but rather, they also affect other non-target species. These work by proliferating through the plant, which can then be consumed by species other than insects, and can also enter waterways, producing downstream effects that are harmful for biodiversity.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: The decomposition of leaves and woody debris is a crucial part of the recycling of nutrients back into the soil. Additionally, fungi and other detritivores derive energy and nutrients from them as well. It can also regulate moisture and soil temperature, as well as provide cover for a variety of forest species.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Sampling was performed via elevated and ground traps, which collected falling debris and leaf litter. The mass of this is then collected, after the debris is separated into a few different classes. Each of these classes are weighed separately. 2. Sampling was performed at various sites of different sizes. These locations are chosen at random, but they all have vegetation of at least 2 meters in height. Within each site, trap placement is either random or targeted, depending on the vegetation within the plot - >50% coverage of plants >2 meters in height, random placement. 3. They sampled the ground traps once per year, but elevated traps received much more frequent sampling, depending on time of year and vegetation. Deciduous forest elevated traps received sampling once every two weeks, while those in evergreen forests were sampled once every 1-2 months. It is noted that the deciduous forests only received sampling at that frequency during senescence.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

```
#> dim(Neonics)
```

```
#[1] 4623 30
```

```
#4623 x 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##           12           102           360           11
##      Cell(s)      Development      Enzyme(s) Feeding behavior
##           9           136           62           255
##      Genetics      Growth      Histology      Hormone(s)
##          82           38           5           1
##      Immunological Intoxication      Morphology      Mortality
##          16           12           22           1493
##      Physiology      Population      Reproduction
##           7           1803           197
```

```
sort(table(Neonics$Effect), decreasing = TRUE)
)
```

```
##
##      Population      Mortality      Behavior Feeding behavior
##          1803          1493          360          255
##      Reproduction      Development      Avoidance      Genetics
##          197          136          102          82
##      Enzyme(s)      Growth      Morphology      Immunological
##           62          38          22          16
##      Accumulation      Intoxication      Biochemistry      Cell(s)
##           12          12          11          9
##      Physiology      Histology      Hormone(s)
##           7          5          1
```

Answer: The most common effects are population (1803), mortality (1493), and behavior (360). Determining the effect of these insecticides on other species of insects, particularly in regards to mortality, is a good representation of the efficacy of the treatment and how it may have undesired effects on non-target species as well. Additionally, there may be some large-scale, population level effects on insect species that are worth exploring.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##           Honey Bee           Parasitic Wasp Buff Tailed Bumblebee
##           667                285                183
## Carniolan Honey Bee           Bumble Bee           Italian Honeybee
##           152                140                113
##           (Other)
##           3083
```

Answer: The six most commonly studied species are the Honey Bee, Parasitic Wasp, Buff Tailed Bumblebee, Carniolan Honey Bee, Bumble Bee, and Italian Honeybee. These are all closely related species, mostly types of bee. What is important about this is that they are pollinators who aid in the fertilization of plants and crops. If they are being adversely affected by the insecticide, that would be important to know.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

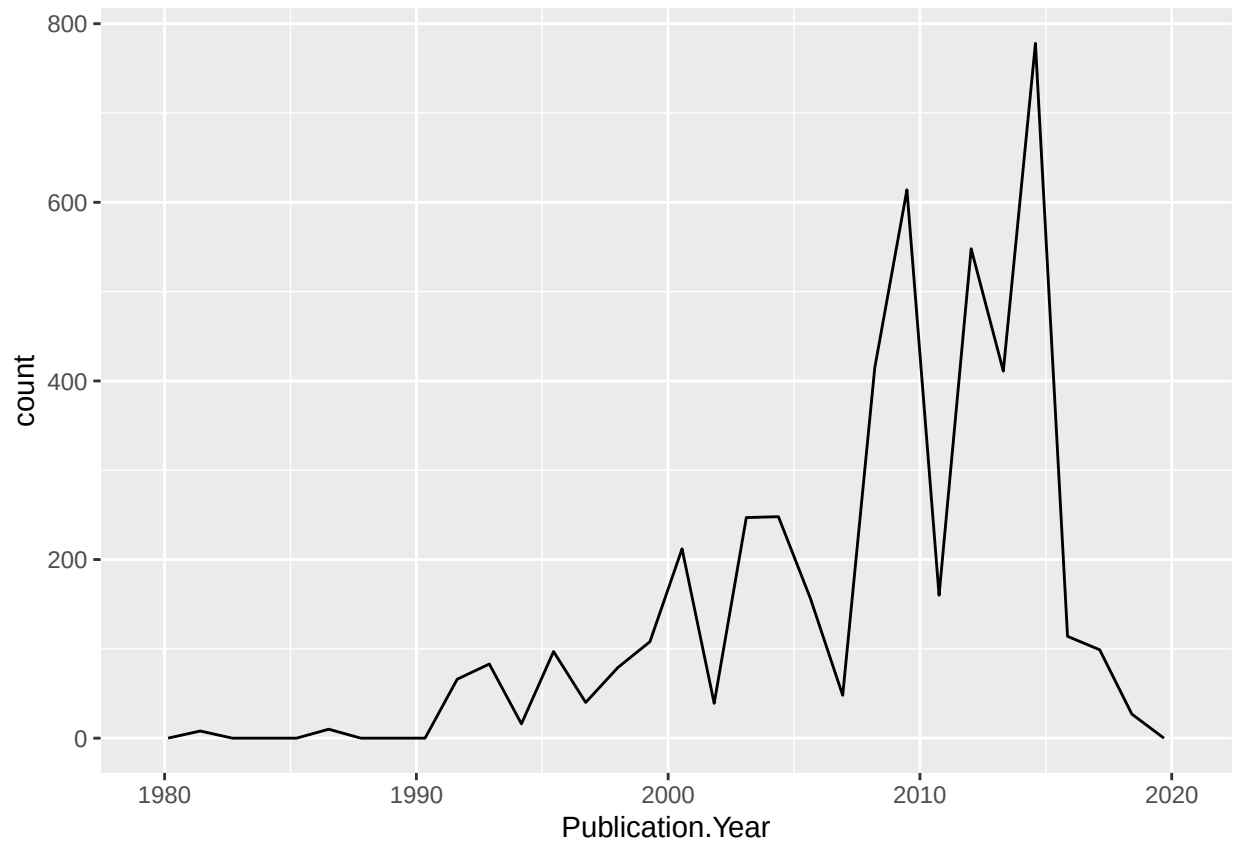
Answer: The class of this column is a factor. It is not numeric because not every result gives a normal number value. For instance, many entries have the result “NR”.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(data = Neonics, aes(x = Publication.Year)) +
  geom_freqpoly()
```

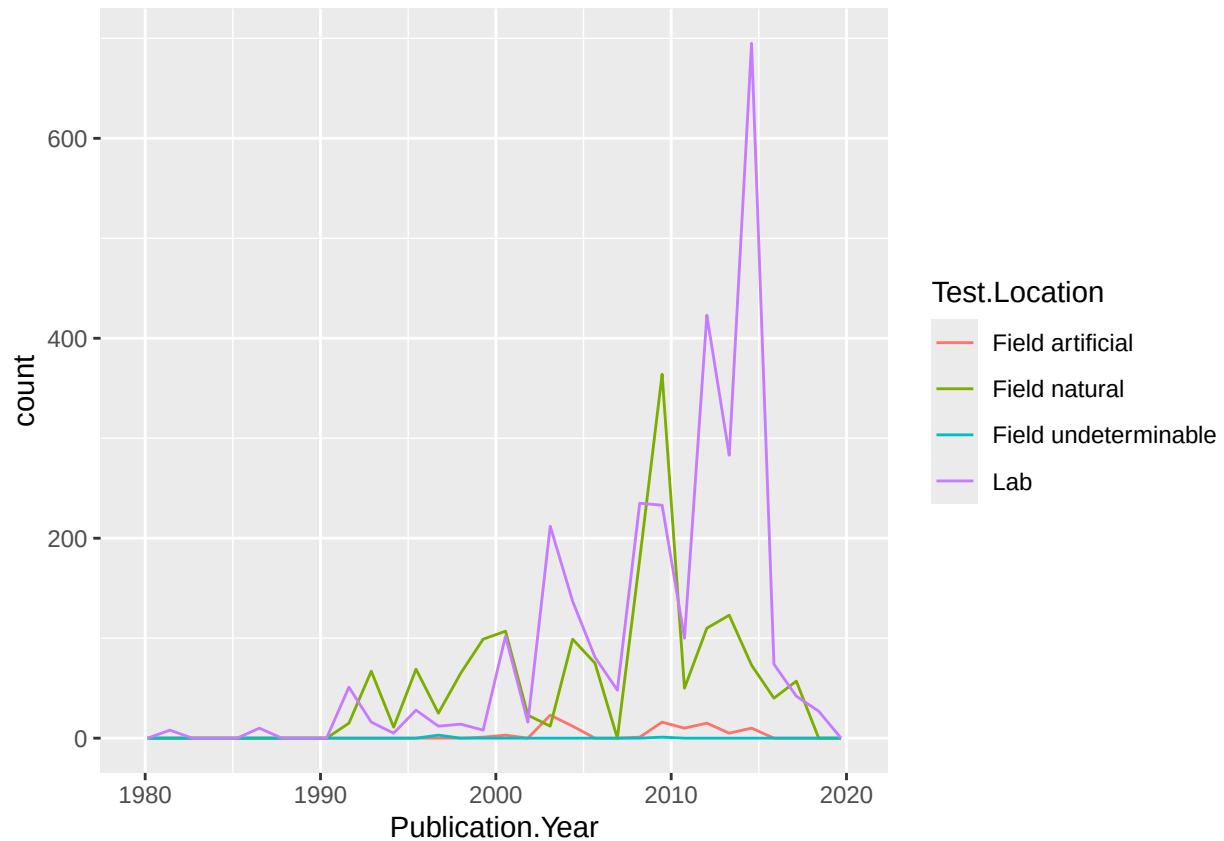
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(data = Neonics, aes(x = Publication.Year, color = Test.Location)) +  
  geom_freqpoly()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



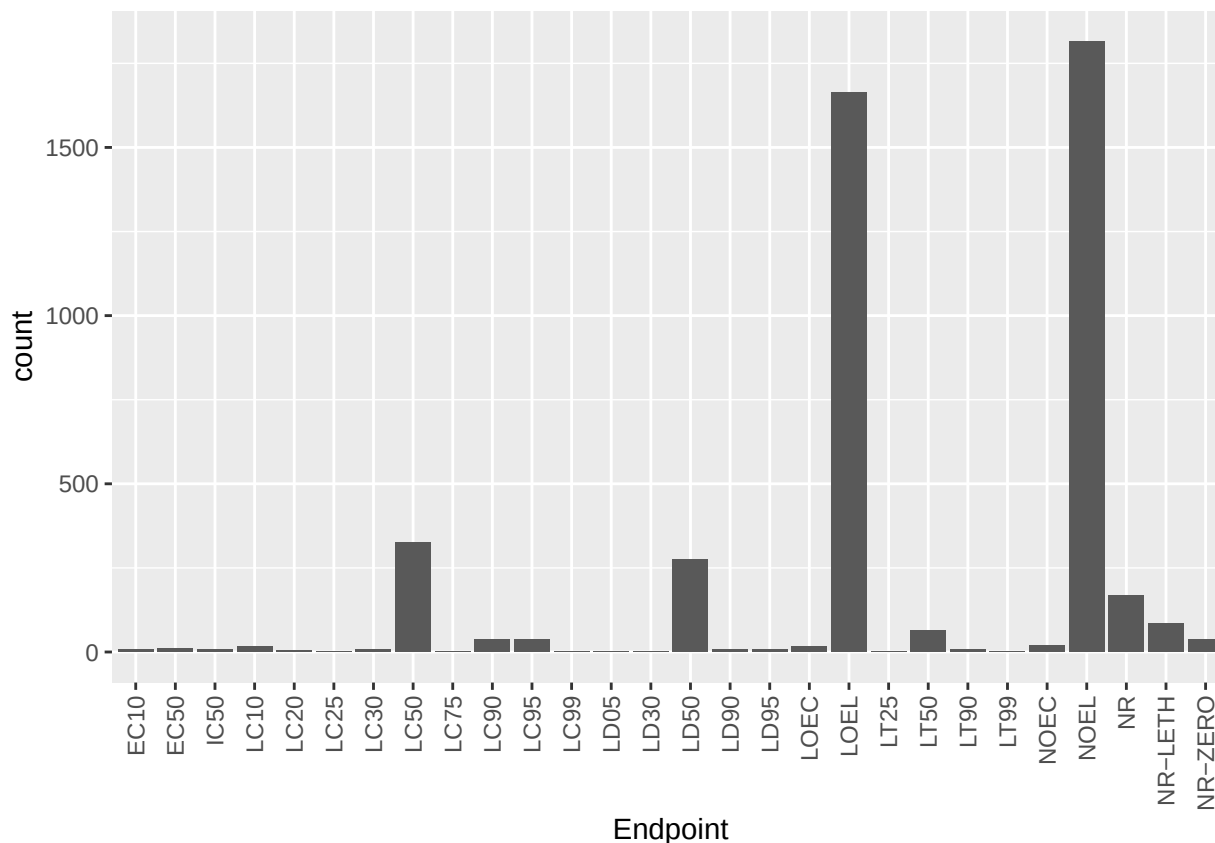
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: Lab and field natural are the most common test locations. Over time, both of these increase, with lab seeing a massive increase between 2010 and 2020.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(data = Neonics, aes(x = Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common end points are NOEL and LOEL. NOEL is defined as no-observable-effect-level, meaning the highest dose that produces effects that fo not differ from the response of the controls. LOEL is defined as the lowest-observable-effect-level, the lowest dose that produces effects that were significantly different from the controls.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#Factor
ymd(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [6] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [11] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [16] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [21] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [26] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
```

```
## [31] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [36] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [41] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [46] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [51] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [56] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [61] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [66] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [71] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [76] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [81] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [86] "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02" "2018-08-02"
## [91] "2018-08-02" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [96] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [101] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [106] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [111] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [116] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [121] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [126] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [131] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [136] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [141] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [146] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [151] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [156] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [161] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [166] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [171] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [176] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [181] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
## [186] "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30" "2018-08-30"
```

```
unique(Litter$collectDate)
```

```
## [1] 2018-08-02 2018-08-30
## Levels: 2018-08-02 2018-08-30
```

```
#2018-08-30 and 2018-08-02
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```
summary(Litter$plotID)
```

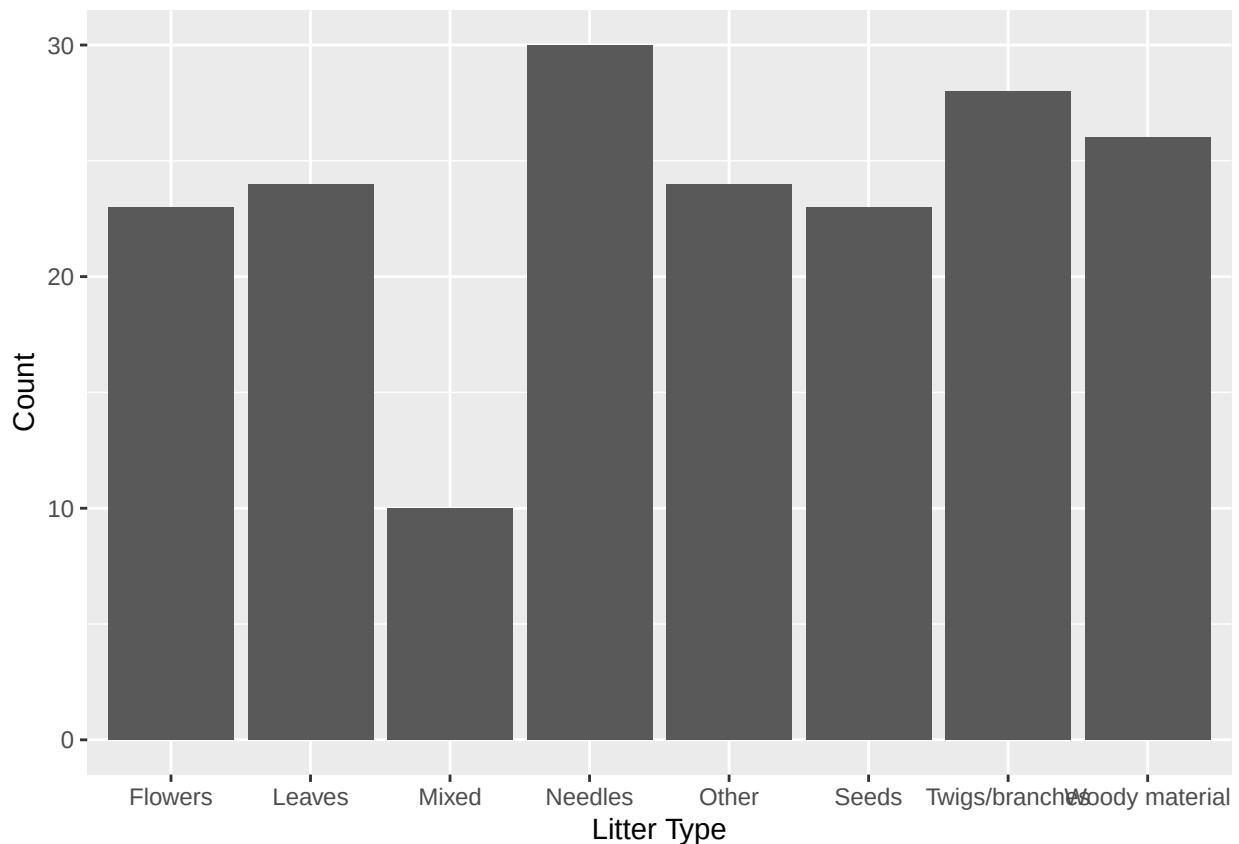


```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14      8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061 NIWO_062 NIWO_063 NIWO_064 NIWO_067 The information obtained from `unique` is different from `summary`, because it only returns the individual values seen in the column of your dataset. It does not list out values more than once, instead it eliminates duplicates and only returns the “unique” values. Summary, in comparison, gives the counts of each of these values, also avoiding duplicates.

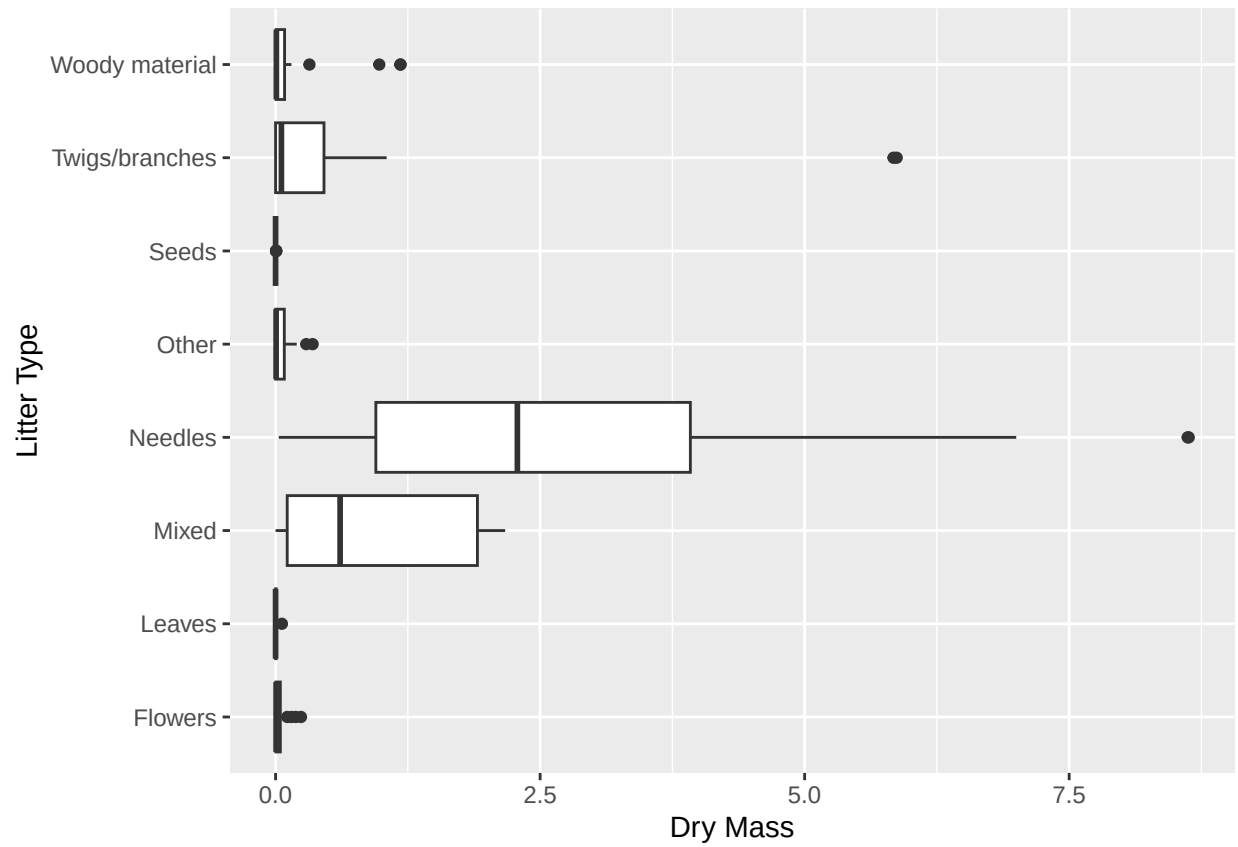
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(data = Litter, aes(x = functionalGroup)) +
  geom_bar() +
  labs(x="Litter Type", y="Count")
```

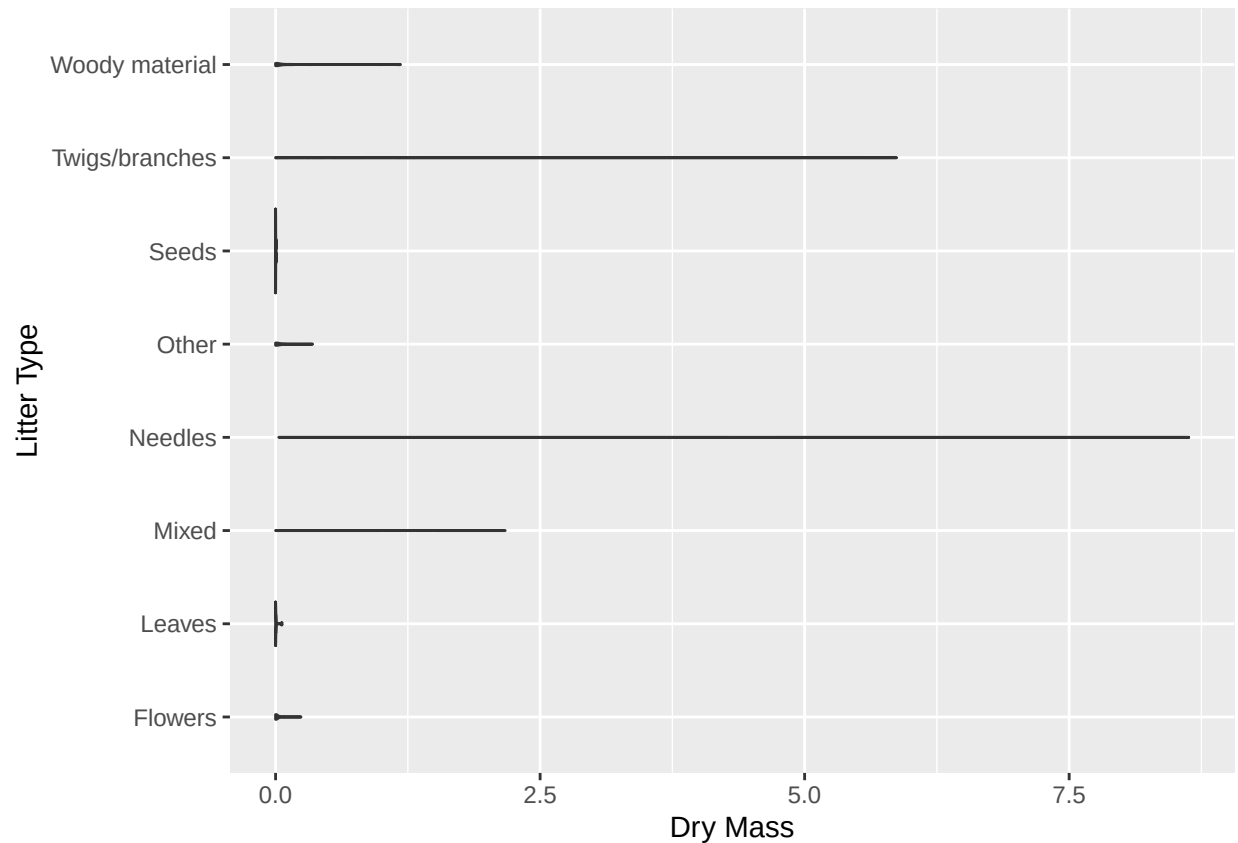


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by functional-Group.

```
ggplot(data = Litter, aes(x=dryMass, y=functionalGroup)) +
  geom_boxplot() +
  labs(x="Dry Mass", y="Litter Type")
```



```
ggplot(data = Litter, aes(x=dryMass, y=functionalGroup)) +  
  geom_violin() +  
  labs(x="Dry Mass", y="Litter Type")
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: In this case, a boxplot is more effective as it showcases quartile values and IQR, as well as outliers. Additionally, the median is shown, which is particularly helpful for analyzing data. A violin plot shows the lowest and highest values and where the values are most concentrated. But for this situation, the densities do not provide much more background to the data.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles have the highest biomass, with mixed litter having the next highest.